

ECN 594: Midterm Exam

Instructions:

- You have **80 minutes**
- You may bring a calculator and notes on a two-sided cheat-sheet (letter-size paper)
- Please be neat. If your work is too messy it will not be graded.
- Be sure to show your working.
- This is a long exam, so there are lots of ways to get points. If you get stuck, move on!
- Good luck!

Name: _____

Question:	1	2	3	4	Total
Points:	30	30	20	20	100
Score:					

1. Short Answer Questions (30 points)

1. For each question, give a number, True/False/NEI (Not Enough Information), or a 1-2 sentence answer.

(a) (4 points) A monopolist faces demand with elasticity $\varepsilon = -2$ and has $MC = \$30$. What is the profit-maximizing price?

(b) (4 points) Consider the logit utility: $u_{ijt} = \beta \cdot \text{sugar}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$. What is the economic interpretation of α ?

(c) (4 points) True, False, or NEI: If we add 5 to the mean utility of all goods (including the outside option), the market shares do not change.

(d) (4 points) Why are BLP instruments (characteristics of other products) *relevant* for price? (1-2 sentences)

(e) (4 points) True, False, or NEI: In the logit model with small market shares, higher-priced products typically have more elastic demand.

(f) (3 points) True, False, or NEI: Under average cost pricing ($P = AC$), deadweight loss is zero.

(g) (4 points) A researcher claims that consumer surplus in logit is $CS = \frac{1}{|\alpha|} (\exp(\delta_1) + \exp(\delta_2))$. Consider a market with two goods ($\delta_1 = \delta_2 = 2$) and an outside option. Is this formula correct?

(h) (3 points) True, False, or NEI: Under perfect price discrimination, deadweight loss is zero.

2. Demand Estimation (30 points)

2. A researcher wants to estimate demand for automobiles using a logit model:

$$u_{ijt} = \beta \cdot \text{mpg}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where mpg_j is fuel efficiency, p_{jt} is price, ξ_{jt} is unobserved product quality, and ε_{ijt} is i.i.d. Type 1 Extreme Value.

- (a) (6 points) Define the mean utility δ_{jt} . Using the Berry inversion, derive an expression for δ_{jt} in terms of observed market shares s_{jt} and the outside option share s_{0t} .
- (b) (6 points) Write down the linear regression equation the researcher would estimate to recover β and α . Clearly identify the dependent variable, regressors, and error term.
- (c) (6 points) A researcher proposes using “Hausman instruments” (prices of the same car in other markets) to address the endogeneity problem. Explain one reason why these instruments might be problematic.

(d) (6 points) Compared to the basic logit, what is one advantage of adding demographic interactions (e.g., $\text{income} \times \text{price}$) to the model?

(e) (6 points) A researcher takes all Toyota Corollas and splits them into two “products” based on the last digit of their license plate: “Corolla-Odd” and “Corolla-Even.” What will the logit model predict about the change in consumer surplus from this split? Should we believe this prediction? Explain.

3. Two-Part Tariffs (20 points)

3. A gym charges a monthly membership fee F and a per-visit price p . Marginal cost per visit is $c = 2$.

Part A: Homogeneous consumers

There are 100 identical consumers, each with demand $q = 20 - 2p$ for gym visits per month.

- (a) (5 points) Write down the gym's profit as a function of F and p . (Hint: What is consumer surplus at price p ? What is the maximum F the gym can charge?)

- (b) (5 points) Show that the profit-maximizing per-visit price is $p^* = c = 2$. (Hint: Take the derivative of profit with respect to p and set it equal to zero. Show your full derivation.)

Part B: Heterogeneous consumers

Now suppose there are 50 “heavy users” with demand $q_H = 10 - p$ and 100 “light users” with demand $q_L = 5 - p$. Marginal cost is $c = 0$.

(c) (10 points) The gym is considering two pricing strategies:

(a) $F = 50, p = 0$

(b) $F = 12.5, p = 0$

Calculate profit under each strategy. Which is better and why?

4. Self-Selection (20 points)

4. A software company sells two versions: Pro (P) and Basic (B). There are two types of users: businesses with high willingness to pay and students with low willingness to pay.

Consumer Type	Willingness to Pay	
	Pro Version	Basic Version
Business (40 users)	\$400	\$150
Student (60 users)	\$120	\$100

Marginal cost is \$30 for both versions.

- (a) (6 points) If the company could perfectly identify consumer types, what prices would it charge for each version to each type? What is total profit?
- (b) (6 points) The company cannot identify types but offers both versions. Write down the incentive compatibility (IC) constraint that ensures businesses buy Pro rather than Basic.

- (c) (8 points) Find the profit-maximizing prices for Pro and Basic under self-selection. (Hint: Which IC constraint binds? Set it as an equality and solve.)